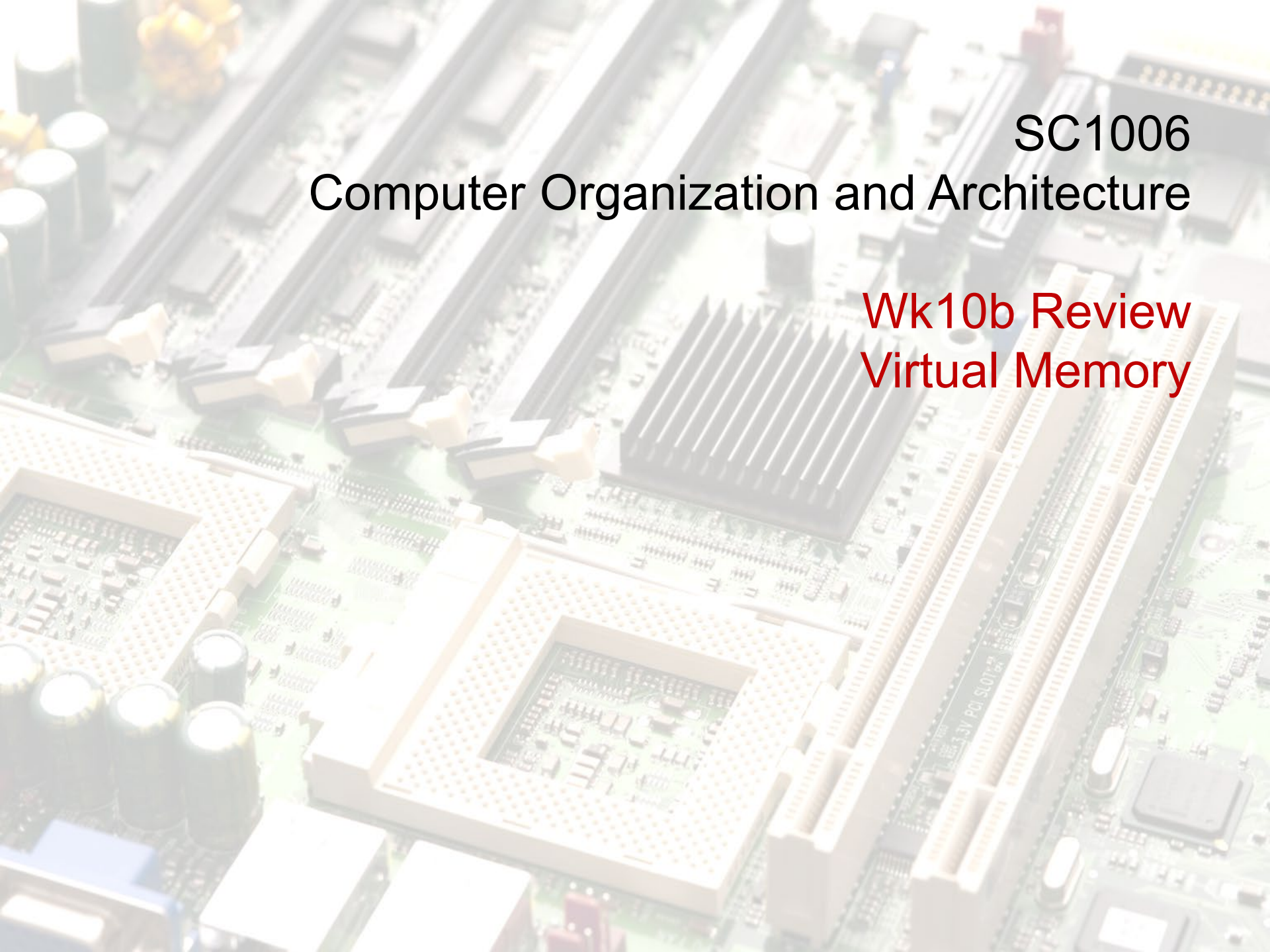




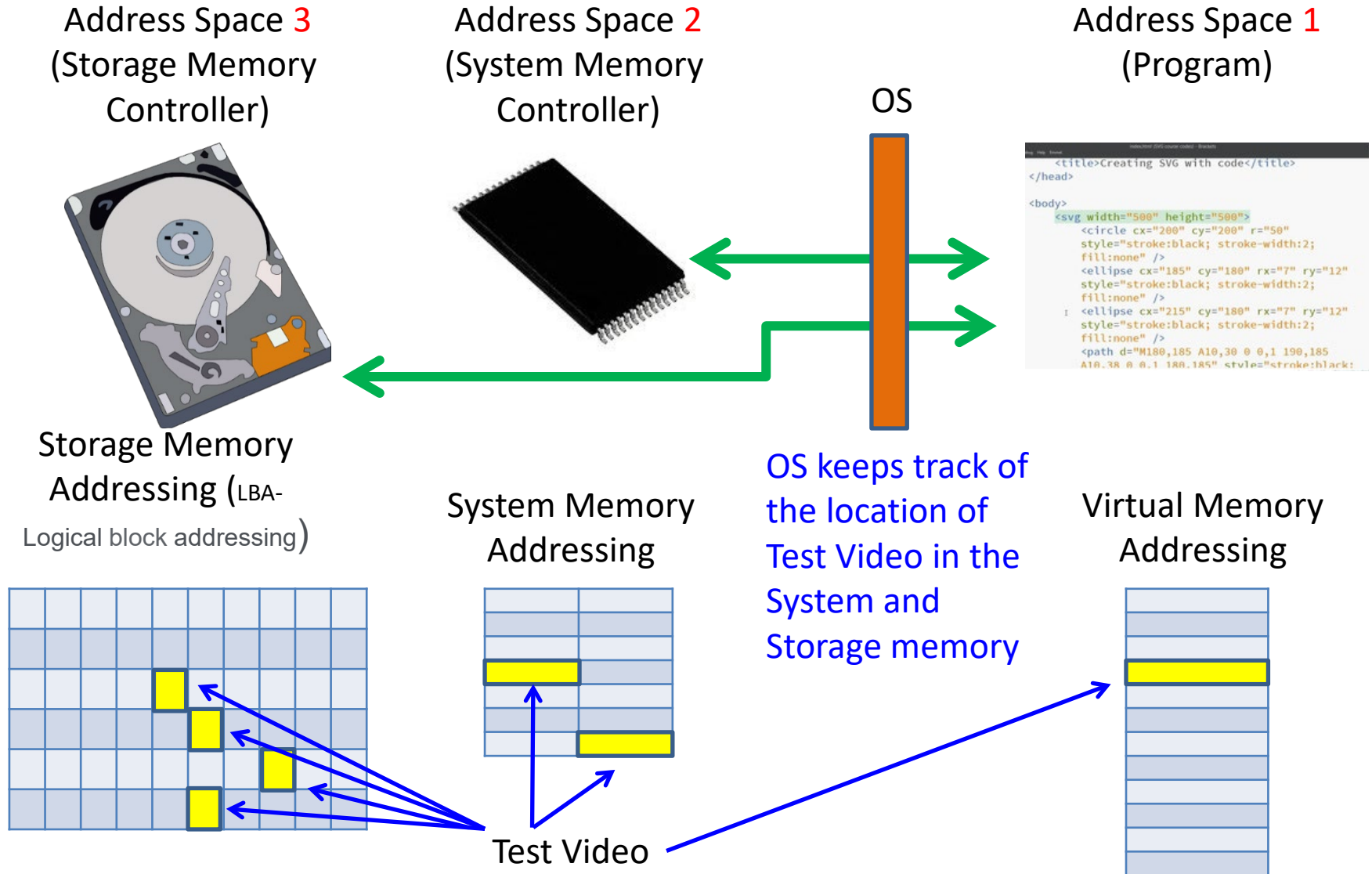
SC1006

Computer Organization and Architecture

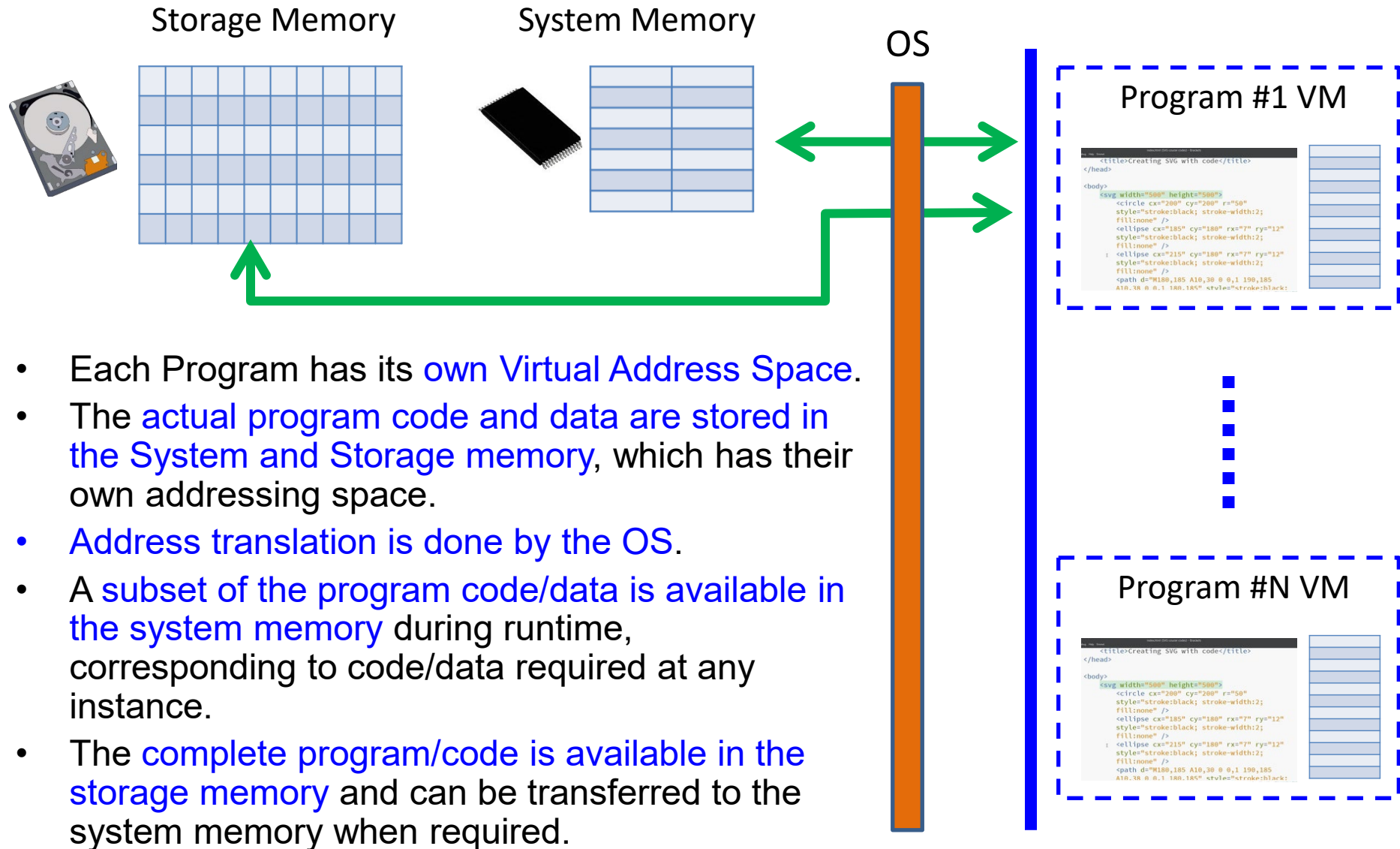
Wk10b Review
Virtual Memory

VIRTUAL MEMORY

Memory Address Space (Computing)

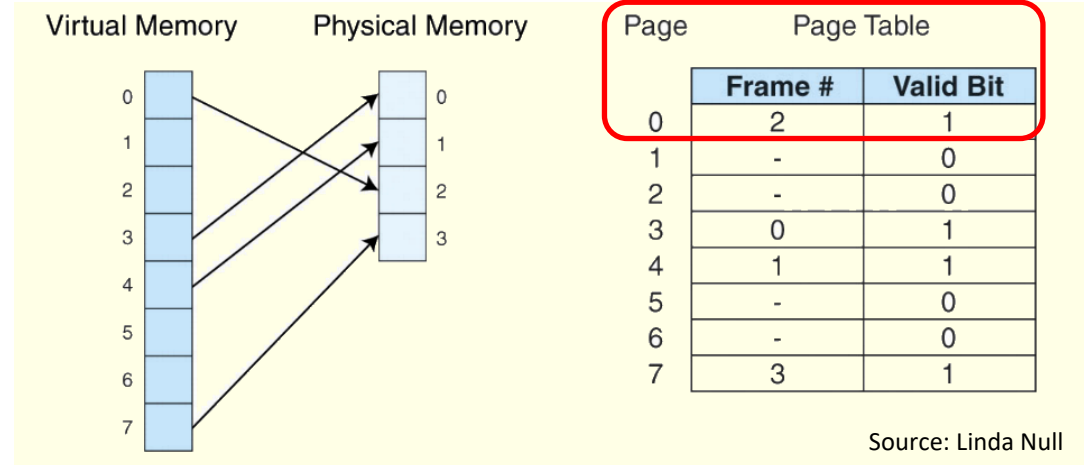
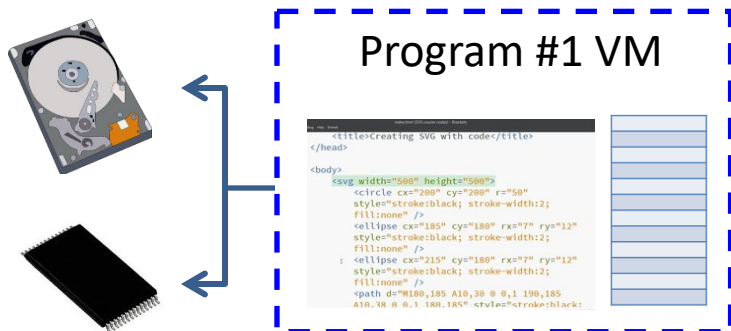


Virtual Memory (VM) Management



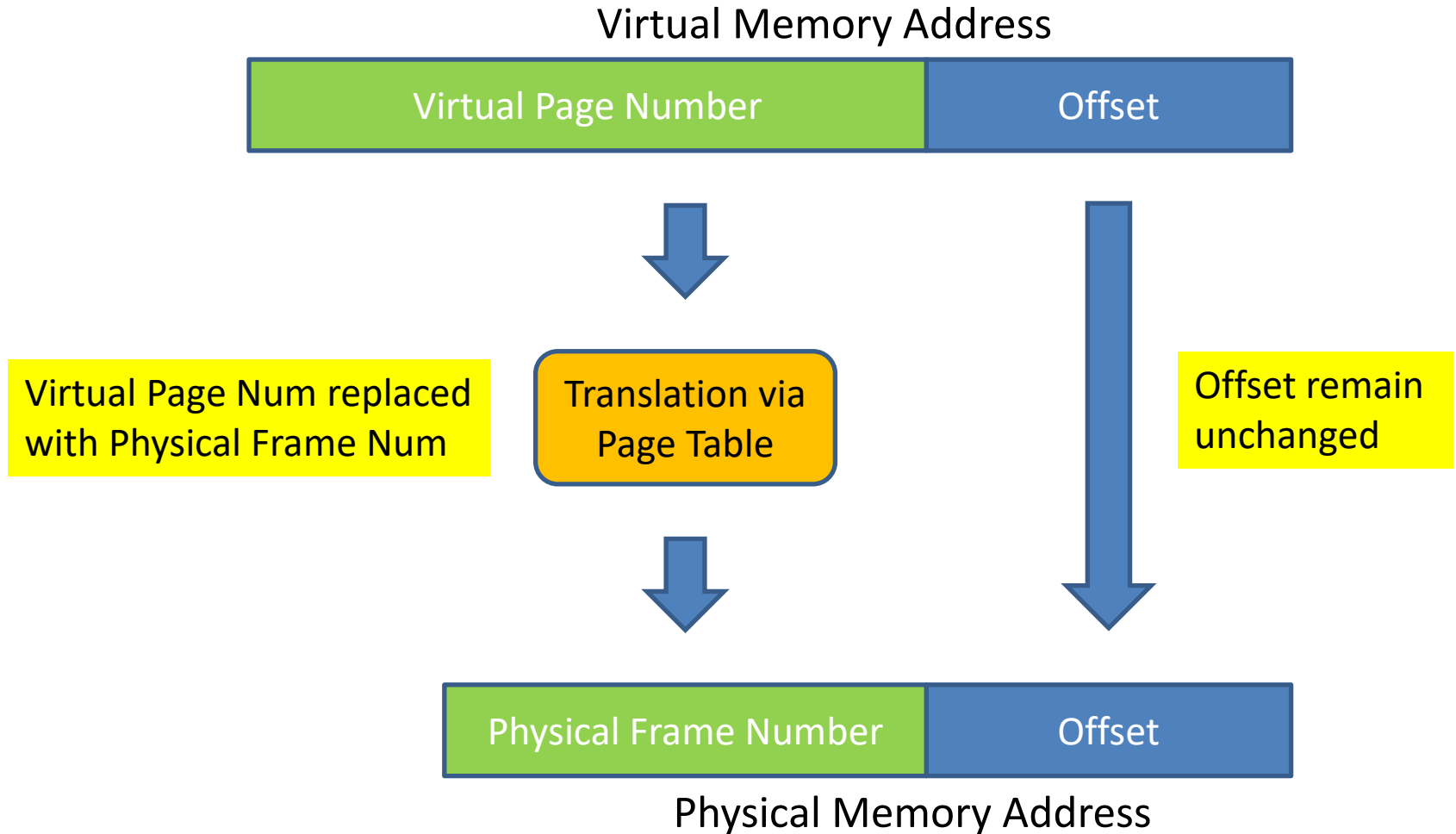
Paging Method

- In a system that uses paging, the memory space is partitioned into **fixed size blocks** known as a **Page/Frame**.
- Information concerning the location of each **page**, whether on disk or in memory, is maintained in a data structure called a **page table** (shown below).
- In the Page Table below
 - **Frame** refers to the **physical frame** number in the main memory.
 - **Page** refers to the **virtual page** number used by program code.
 - **Valid Bit (VB)** indicates whether the Virtual Page is in the main memory (VB=1) or not (VB=0).



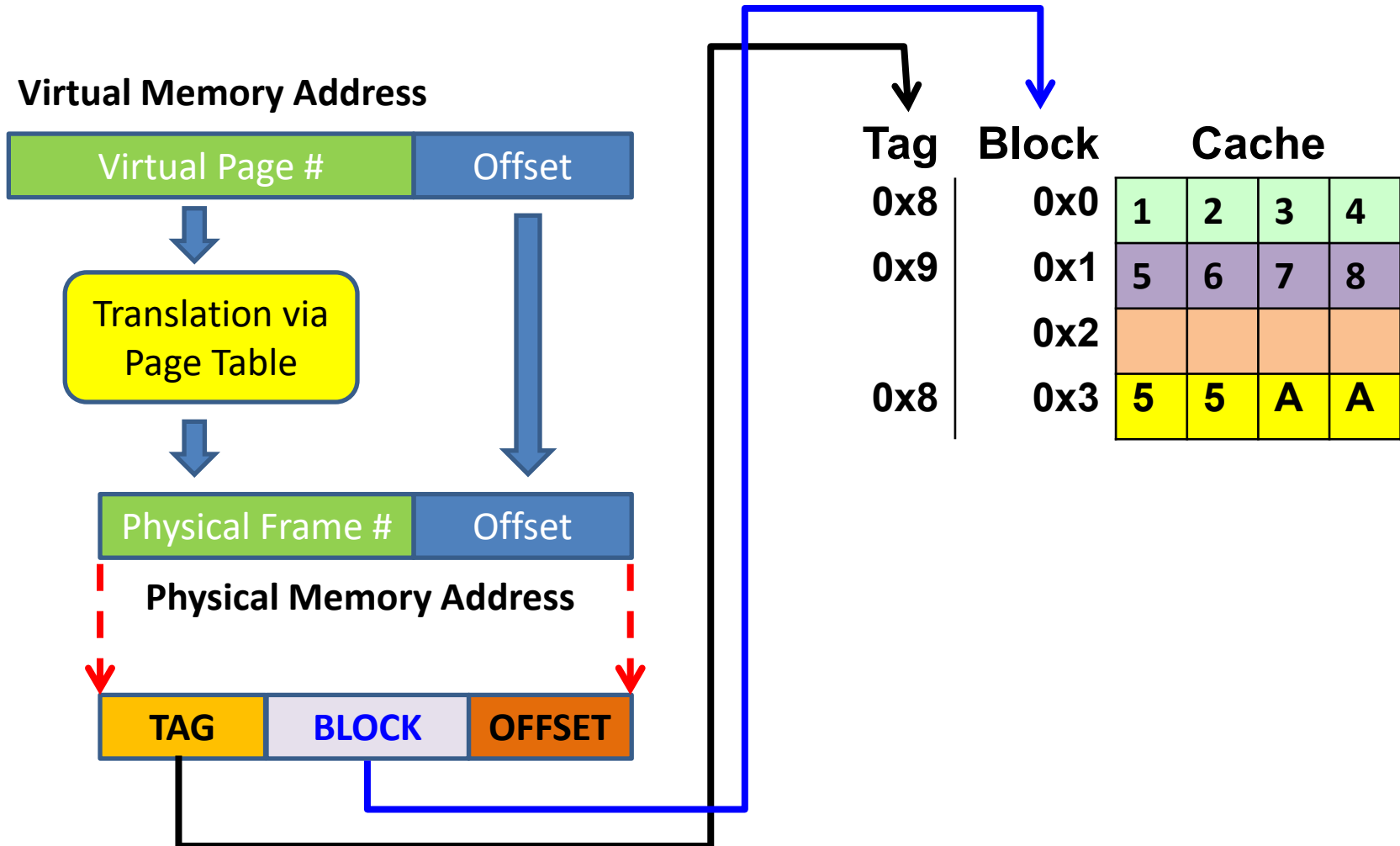
Address Translation

Work in Binary Numbers format



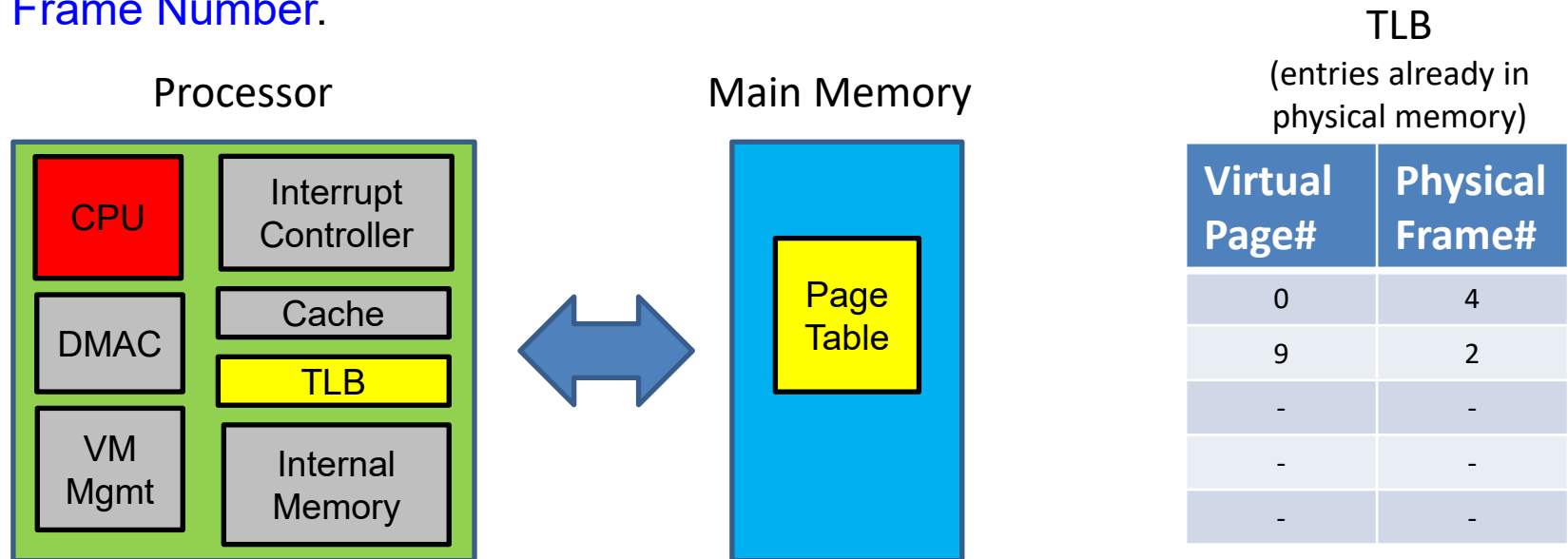
Integrating Cache and Virtual Memory

- Assumption: Cache uses Physical Memory Address to perform Mapping.

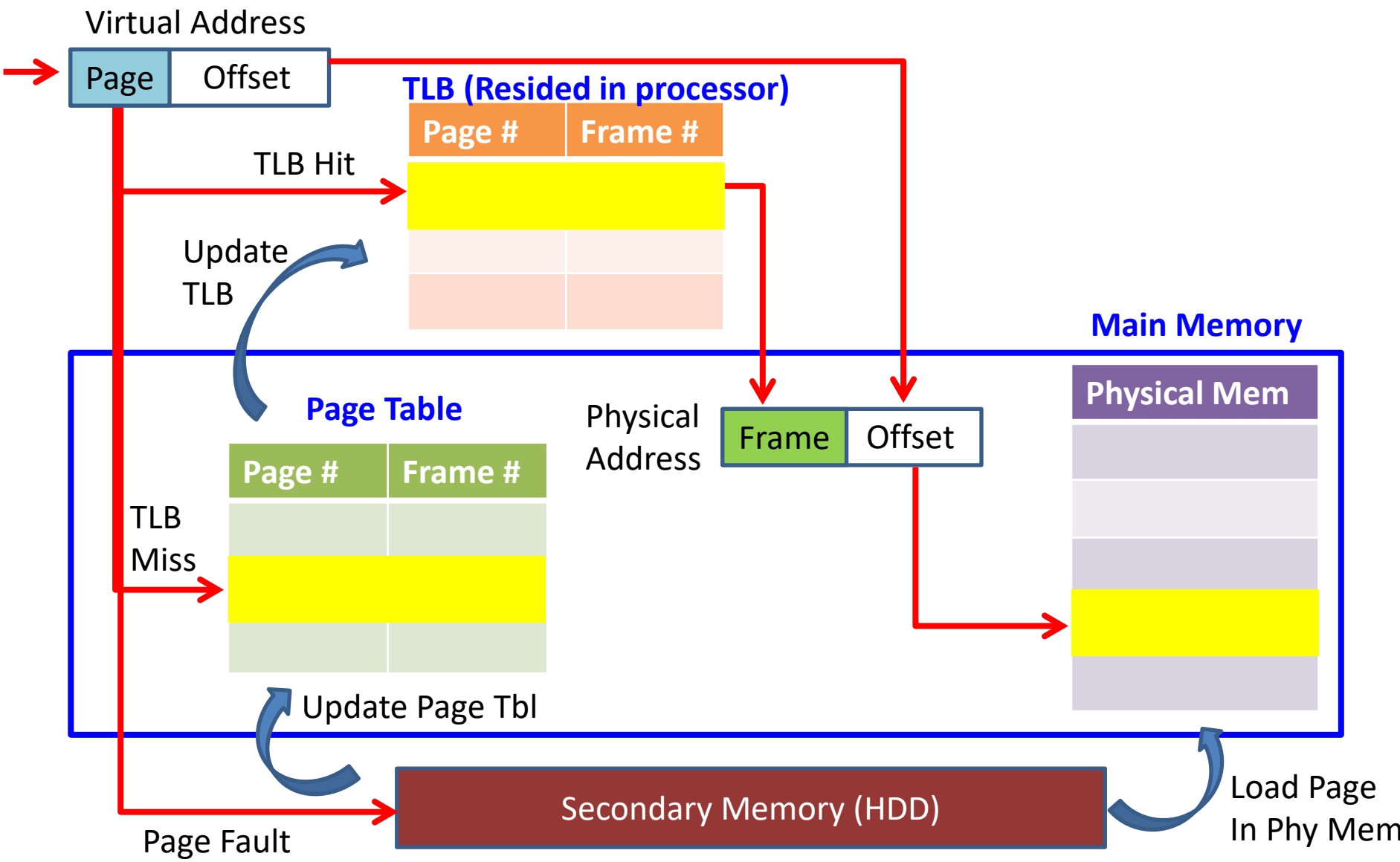


Translation Lookaside Buffer (TLB)

- **Page Table** is located in **Main Memory** so access is **relatively slower** than internal memory access.
- To **speed up** the page translation process, a **page table cache (TLB)** is used to store a list of most recently/frequently used address translation entries in the page table.
- TLB is implemented with **fast memory** such as SRAM and resides **within the processor**.
- Each **TLB entry** includes a **Virtual Page number** and its corresponding **Frame Number**.



TLB Lookup Process



Cache mapping & tracking



In a system with 32 virtual pages and 8 physical frames, what is the maximum number of valid entries in the page table?

Page table contains entries for every Virtual Pages; but not all are valid.
Only virtual pages that are mapped to Physical Frames are valid.

Therefore, the answer: 8

In a system using paging scheme with 32 virtual pages, 8 physical frames and virtual address range of 64KByte, what is the size of one physical frame?

64KByte divided into 32 virtual pages

⇒ Each page = $64\text{KByte}/32 = 2 \text{ Kbyte}$

⇒ Phy Frame size = Virtual Page Size = 2KByte

In a system with 1 Kbyte page size, if the virtual page 0 is mapped to physical frame 3, what physical address will virtual address 0x300 be translated to?

1 Kbyte => 10 bits offset

virtual address 0x300 = **00** 11 0000 0000

Virtual page 0 → Phy Frame 3

⇒ **11** 11 0000 0000

So, phy address: 0x0F00

thinking questions toward understanding...

(some of you asked)

What is the difference of baud rate
& data rate?



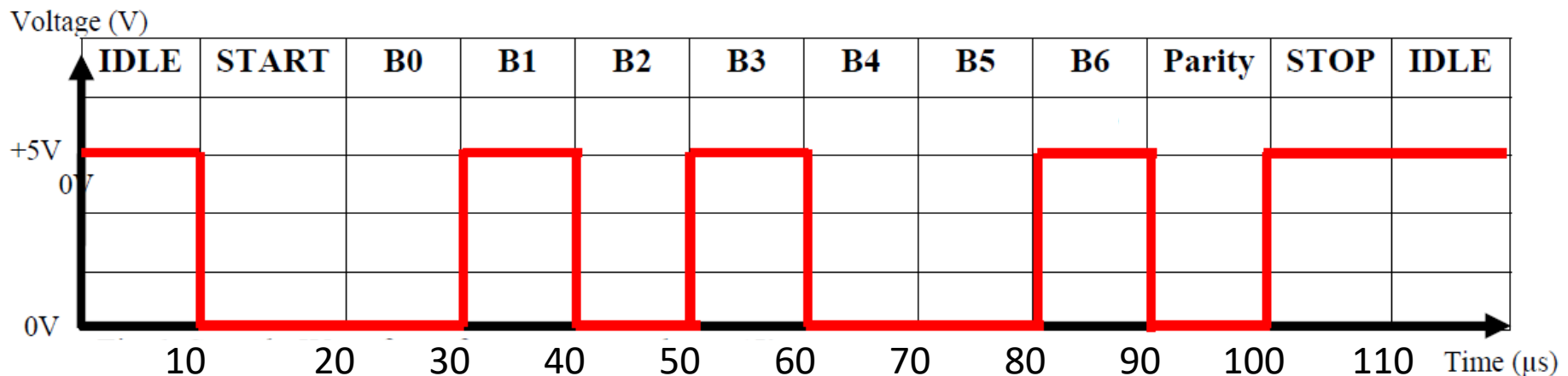
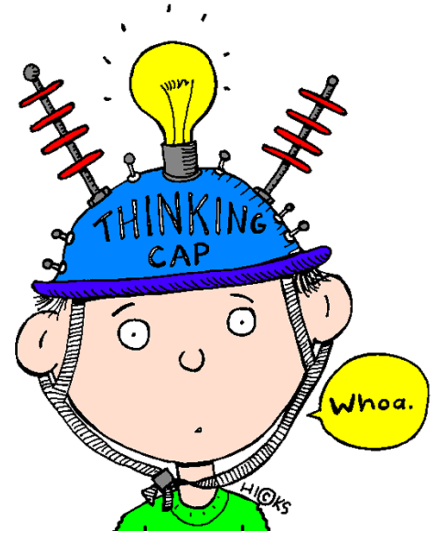
thinking questions toward understanding...

(some of you asked)

What is the difference of data rate & baud rate?

Data rate: actual **data** transmitted per second- B0 ~ B6

Baud rate: all **signal** changes (or symbols) transmitted per second- B0 ~ B6, Idle, Start, Parity, Stop.



thinking questions toward understanding...

How do you summarize the reason(s) why virtual memory is needed?



thinking questions toward understanding...

How do you summarize the reason(s) why virtual memory is needed?

- Allowing systems to utilize/overcommit more memory than physical one
- Convenience, Isolation and Security: each process operates in its own virtual address space, as one process cannot directly access or corrupt the memory of another.
- Efficient Memory Utilization: to optimize memory usage by loading only the necessary parts of programs into physical memory
- Simplified Memory Management: Programmers can work with a consistent virtual memory model without needing to manage physical memory allocation. The OS handles address translation
- Facilitating better multitasking: each task with its own memory space, without interference.



Review & Q/A sessions

Week	Mon	Fri
8 (07/10-11/10)	Review (on-site)	Q/A (on-line)
9 (14/10-18/10)	Review (on-site)	Review (on-site)
10 (21/10-25/10)	Review (on-site)	Review (on-site)
11 (28/10-01/11)	Q/A (on-line)	Q/A (on-site/line)
12 (4/11-08/11)	Review (on-site)	Review (on-site)
13 (11/11-15/11)	Review (on-site)	Q/A (on-site/line)

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$$j_j \in \mathbb{R}^n, p \in \mathbb{R}^n, b'' \in \mathbb{R}^n, 1 \leq j \leq n$$

I] $\tilde{a}\tilde{a}f \sim h_j p, \dots \sim$

$$\geq \frac{1}{2} \|\bar{\mathbf{V}}_y\|^2.$$

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